WORKED SOLUTION

STEP III 2018, Question 10

Mechanics • Compound Pendulum

STEP III 2018, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/18-s3-q10

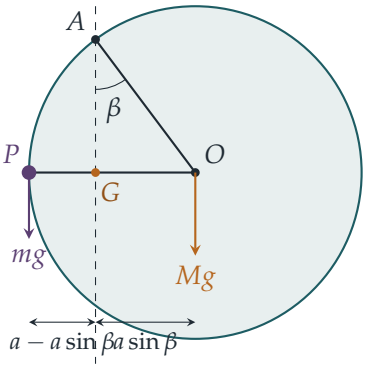
The key idea

Disc and particle together form a single rigid body turning about the fixed axis L, so everything is controlled by one angle. In equilibrium the combined centre of mass must lie directly below A, which is a moment balance about A. For the motion, energy is conserved because the axis is smooth and the reaction there does no work. Once the energy is written as a function of θ and $\dot{\theta}$, differentiating it gives the equation of motion, and small oscillations follow by replacing $\sin\theta$ with θ.

Solution

We model the disc as a uniform rigid lamina, the particle as a point mass fixed to its rim, and the axis L as smooth.

Equilibrium. The only forces with a moment about A are the two weights: Mg acting at O and mg acting at P. The reaction at A passes through A. Since AO makes angle β with the downward vertical, O is a horizontal distance $a\sin\beta$ to one side of A, say to the right. The weight Mg then turns the disc clockwise about A, and only mg can balance it, so P must be on the *left* of the vertical through A. Because OP is horizontal and has length a, the horizontal distance of P to the left of A is $a - a\sin\beta$. (Here $a\sin\beta < a$, so P really is on the left of the vertical.)



Taking moments about A,

$$Mg\,a\sin\beta = mg\,(a - a\sin\beta) \quad\Longrightarrow\quad \sin\beta = \frac{m}{M+m}.$$

Both O and P are a vertical distance $a\cos\beta$ below A. So, in the triangle with AP as hypotenuse, the horizontal side is $a(1-\sin\beta)$ and the vertical side is $a\cos\beta$:

$$AP^2 = a^2(1-\sin\beta)^2 + a^2\cos^2\beta = a^2(2-2\sin\beta)$$
$$= 2a^2\Big(1 - \frac{m}{M+m}\Big) = \frac{2Ma^2}{M+m},$$

so $AP/a = \sqrt{2M/(M+m)}$, as required.

It's worth noting where the combined centre of mass G is. It lies on the segment OP, because it is a weighted average of O and P, so it is at depth $a\cos\beta$ below A. In equilibrium it is directly below A, so

$$AG = a\cos\beta.$$

Conservation of energy. The disc and particle move as one rigid body. In equilibrium OP is horizontal, and OP turns with the body, so when OP makes angle θ with the horizontal, the whole body has turned through θ from its equilibrium position. Its angular speed is therefore $|\dot\theta|$.

Ask yourself How fast is the particle moving?

Every point of the body moves on a circle centred on the axis. The particle stays a fixed distance AP from A, so its speed is $AP\,|\dot\theta|$ and its kinetic energy is

$$\tfrac12 m\,AP^2\,\dot\theta^2 = \tfrac12 m\cdot 2a^2(1-\sin\beta)\,\dot\theta^2 = (1-\sin\beta)\,ma^2\dot\theta^2,$$

using $AP^2 = 2a^2(1-\sin\beta)$ from above. The disc's kinetic energy is $\frac12 I\dot\theta^2$.

For the potential energy, treat the total mass $M+m$ as concentrated at G. The line AG is vertical in equilibrium, and it has also turned through θ, so G is now a depth $a\cos\beta\cos\theta$ below A. Measuring potential energy from the equilibrium position,

$$\text{PE} = (M+m)g\big(a\cos\beta - a\cos\beta\cos\theta\big) = (M+m)ga\cos\beta\,(1-\cos\theta).$$

The axis is smooth and the reaction at A acts at a point that doesn't move, so it does no work. Gravity is the only other force, and it is conservative. So kinetic plus potential energy is constant:

$$\tfrac{1}{2}I\dot\theta^2 + (1-\sin\beta)\,ma^2\dot\theta^2 + (m+M)ga\cos\beta\,(1-\cos\theta) = \text{constant},$$

which is the required result.

Small oscillations. Differentiating with respect to t:

$$\dot\theta\Big[\big(I + 2(1-\sin\beta)ma^2\big)\ddot\theta + (M+m)ga\cos\beta\,\sin\theta\Big] = 0.$$

The bracket must vanish whenever $\dot\theta \neq 0$, and by continuity it vanishes at every instant. For small θ, $\sin\theta \approx \theta$, so

$$\ddot\theta = -\omega^2\theta, \qquad \omega^2 = \frac{(M+m)ga\cos\beta}{I + 2(1-\sin\beta)ma^2},$$

which is simple harmonic motion with period $2\pi/\omega$.

Now put $m = \tfrac{3}{2}M$. Then $\sin\beta = \dfrac{3/2}{5/2} = \tfrac{3}{5}$, and since β is acute, $\cos\beta = \tfrac{4}{5}$. The numerator is

$$\tfrac{5}{2}M \cdot ga \cdot \tfrac{4}{5} = 2Mga,$$

and with $I = \tfrac{3}{2}Ma^2$ the denominator is

$$\tfrac{3}{2}Ma^2 + 2\cdot\tfrac{2}{5}\cdot\tfrac{3}{2}Ma^2 = \tfrac{3}{2}Ma^2 + \tfrac{6}{5}Ma^2 = \tfrac{27}{10}Ma^2.$$

So $\omega^2 = \dfrac{20g}{27a}$, and the period is

$$\frac{2\pi}{\omega} = 2\pi\sqrt{\frac{27a}{20g}} = 2\pi\cdot\frac{3}{2}\sqrt{\frac{3a}{5g}} = 3\pi\sqrt{\frac{3a}{5g}},$$

using $\sqrt{27/20} = \sqrt{9\cdot 3/(4\cdot 5)} = \tfrac{3}{2}\sqrt{3/5}$. This is the required period.

Checks

- With $m = 0$ we get $\sin\beta = 0$ and $AP/a = \sqrt{2}$. The disc hangs with O below A, and a rim point level with O is indeed $\sqrt{2}\,a$ from A. The period becomes $2\pi\sqrt{(3a/2)/g}$, the familiar result for a disc pivoted at its rim.
- The kinetic-energy terms add to $\tfrac{1}{2}(I + m\,AP^2)\dot\theta^2$, the moment of inertia of the whole body about L, as they should.
- A step-by-step computer simulation of the rigid body, with $m = \tfrac{3}{2}M$ and $a = 0.5\,\text{m}$, gave a small-oscillation period of $1.64815\,\text{s}$. The formula gives $3\pi\sqrt{1.5/49.05} \approx 1.64815\,\text{s}$. The simulation also showed that the energy expression stays constant for large swings.

Common slips

- Putting P on the same side of the vertical as O. Then the two weights turn the body the same way and there can be no equilibrium.
- Using a instead of AP for the particle's distance from the axis. The particle moves on a circle about A, not about O.
- Finding the potential energy of O and P separately and getting lost in the trigonometry. Using the combined centre of mass G, at distance $a\cos\beta$ from A, is much cleaner.

Beyond STEP

This is a *compound pendulum*: any rigid body swinging about a fixed axis oscillates like a simple pendulum of length $I_{\text{total}}/\big((M+m)h\big)$, where h is the distance from the axis to the centre of mass. Our energy method is the one-variable case of Lagrangian mechanics, in which the equations of motion come from the kinetic and potential energies written in a single generalised coordinate. Kleppner and Kolenkow's *An Introduction to Mechanics* treats rigid-body rotation and the physical pendulum carefully.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.